

CS 111: Due Date Apr 15, 2026 (Wednesday), 9am

1. What is the difference between white noise and salt and pepper noise?
2. In the Gaussian pyramid, why is it that we can reduce the size of the image (as we go through different levels) without violating the Nyquist sampling rate?
3. Show that bilinear interpolation is not a linear function?
4. Why can we combine the blurring filter and the gradient filter into one filter for edge detection?
5. We are given an 128x128 image (indexed by 0 to 127) which has 3 vertical lines, each 1 pixel width at columns at column number 32, 64, and 96. I want to reduce the size of the image to 64 by 64 and pick odd pixels in horizontal and vertical direction to achieve the goal.
 - a. What image will I get? Would it be a faithful representation of the original 128x128 image?
 - b. How can you explain the phenomenon you see? What is it called?
6. Which process in the Canny edge detector helps to maintain continuity of edges? Explain your answer.
7. Why are Canny edges usually thinner than those by Sobel or Laplacian operators?
8. A parabola is given by the equation $y^2 = 4ax$. I want to detect parabolas in an image using Hough transform.
 - a. What is the dimension of the Hough space?
 - b. What geometric entity does a point in the image space correspond to in the Hough space?

1. White Noises is random and small disturbance. They are uniformly distributed around the original signals.

Salt and Pepper Noise is a type of impulse noise which is sparse but has strong disturbance.

2. Before we reduce the size of image, we use Gaussian filter, which removes high-frequency components and reduces the signal bandwidth, to downsample the image.

3. We do interpolation in both horizontal and vertical direction

$$A = I(i, j)(1-a) + I(i+1, j)a$$

$$B = I(i, j+1)(1-a) + I(i+1, j+1)a$$

Then do interpolation in z direction, obtain $C = I(i, j)(1-a)(1-b) + I(i+1, j)a(1-b) + I(i, j+1)(1-a)b + I(i+1, j+1)ab$

We have term " ab ", the function is not linear

4. Convolution is associative: $a * (b * c) = (a * b) * c$

So $H_{grad} * (H_{blur} * I) = (H_{grad} * H_{blur}) * I$, then the blurring filter combines gradient filter and do not influence the result.

5. a) If we downsample by selecting only odd-indexed pixels, the three vertical lines (32, 64, 96) will not be sampled and shown in result. It is not faithful.

b) Because we didn't do prior low-pass filtering, which causes high-frequency components to be mixed. It is called Aliasing.

6. The process is called Hysteresis. Weak edges are preserved when they are connected to strong edges. This helps keep whole structure.

7. It is the non-maxima suppression. It checks marked pixels and retain those whose value is highest amongst all the marked 8-connected neighbours.

8. (a) For $y^2 = 4ax$, then $a = \frac{y^2}{4x}$, so in Hough Space is 1-dimensional

(b) For (x_0, y_0) in Image space, then $a = \frac{y_0^2}{4x_0^2}$ is a single point.

Thus a single point in the image space corresponds to a single point in Hough space.